OWNER : **INDIAN OIL CORPORATION LIMITED (IOCL)**

PMC : **TOYO ENGINEERING INDIA PRIVATE LIMITED (TEIPL)**

LSTK CONTRACTOR : **L&T HYDROCARBON ENGINEERING LIMITED (LTHE)**

UNIT / UNIT NO : **TEG BARRELLING FACILITY/207D**

PMC JOB NO. / NAME : **6389 / IPEG**

LTHE JOB NO. : **IOCL-O&U**

Drawing/Document Category (Submission Purpose)	**Drawing/Document Review Status (By OWNER / PMC)**		
☐ For Approval ☐ For Review ☐ For Information	**Approval Code**	**Status**	**Action Required**
	1	Rejected	Revise & resubmit
	2	Commented as noted	Work may proceed subject to acceptance & incorporation of comments
	3	Acceptable with minor comments	Incorporate comments, if any. Resubmit as certified final. Work may proceed.
	4	Approved with No comments-FINAL	Work may proceed.
	5	Information Only	Accepted for information only. Work may proceed.
	NAME	**DISCIPLINE**	**SIGN.** / **DATE**
	This Approval does not absolve the CONTRACTOR from meeting their contractual obligations of delivering a complete safe and efficiently operating unit facilities meeting all requirements of the Contract.		

I1	Issued for Inquiry	AC	NSH	PSS	-	RMK	19-July-19
Rev.	**Description**	**Prepared By**	**Reviewed By**	**Approved By**	**Reviewed By**	**Approved By**	**Approved Date**

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CHANGE LIST

SL. NO.	REV. NO.	PAGE NO.	CLAUSE / LINE NO.	BRIEF DESCRIPTION / REASON OF REVISION	REMARKS
1	I1	-	-	Issued for Inquiry	-

A Revision line in left side margin denotes revision

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PROJECT BRIEF

M/s Larsen & Toubro Hydrocarbon Engg. Ltd. (LTHE), has been awarded the LSTK contract for installation of Ethylene Glycol production facility comprising of an ERU (Ethylene Recovery unit) of capacity 180 KTA, Ethylene Glycol unit of capacity 357 KTA (comprising of MEG – 332 KTA, DEG-24 KTA and TEG-1 KTA), Intermediate and Product Storage, Dispatch facilities & Utility facilities at Paradip Refinery Complex.

Toyo Engineering India Pvt. Ltd. (TEIPL) has been selected by Owner as their PMC for this project.

Project has been separated with three LSTKs. The details of LSTKs are as follows
LSTK – Package – 1 – Ethylene Recovery Unit & Ethylene Glycol Unit
LSTK – Package – 2 – Off site Facilities & Utilities
LSTK – Package – 3 –MEG Dispatch Facilities, Jetty and Associated Work

This document shall cover scope related to LSTK Package – 2 only.

The Ethylene Glycol project consists of Ethylene Recovery Unit (ERU), Ethylene Glycol (EG) Unit, Offsite and utilities for the project. The units can be tabulated as follows.

Sr. No.	**Facility Capacity**	**Licensor**
Ethylene Recovery Unit (ERU)	180 KTPA	McDermott International, USA
Ethylene Glycol Unit (EG) MEG DEG TEG	 332 KTPA 24 KTPA 1.06 KTPA	Scientific Design (SD), USA
Offsite and Utility facilities	**As required for Ethylene Glycol project based on inputs from Process Licensors**	**By LSTK-2 Contractor**

L&T Chiyoda Ltd. (LTC), Vadodara is carrying out the detail engineering activities for the project.

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1. ITEMS TO BE QUOTED

The following items (material and services), required for Indian Oil Corporation Limited (IOCL) at Paradip Refinery Complex, Odisha, India, shall be supplied in accordance with the requirement specified in this requisition and documents attached herewith.

Sl. No.	Description	Requirement	Scope of supply & work
1	Capacity	6 Ton	By Supplier
2	Height Adjustment	Above Dock Operation : 600 mm, Below Dock Operation : 300 mm.	By Supplier
3	Quantity of dock leveler	One (1)	By Supplier
4	Platform Size	As per supplier	By Supplier
5	Electrical equipment including motor	1) Supplier to Consider Safe Area for Electrical Equipment. Motor shall be IP-55 minimum. 2) Motor Energy efficiency – IE2 3) Vendor to note & confirm that Motors 0.18Kw – 160 kW: 415V, +/- 10%. 3 Phase, 50Hz, +3 to -5%. 4) And For Motors < 0.18 kW: 240V, +/- 10%. 1 Phase, 50Hz, +3 to -5%. 5) Starting locked rotor current for 415V LV motors : 6.5 times full load current without any upward tolerance 6) For Motor, Terminal box preferably be phase segregated type and shall be suitable to withstand fault level "50kA for 0.25 sec" 7) Motor Terminal box shall be rotatable by 360 Degree in steps of 90 Degree. 8) Motors shall be Class F insulated & temperature rise shall be limited up to Class B.	By Supplier
6	Local Control Panel / Local Control Station for Dock leveler	LTHE will provide only single point power supply (Power feeder) to Control Panel of Dock Leveler, Further distribution including Motor starter, Cables, Lugs, Glands shall be in Supplier's scope. Any Local control station/ Start stop push button required shall be in Supplier's scope only	By Supplier
7	Earthing Requirement	Supplier to provide suitable arrangement/ earthing studs and earthing boss for connecting to earthing grid by purchaser.	By Purchaser
8	Field Instruments & controls (If Applicable)	All Instruments shall be suitable to Hazardous area Zone 1, Group IIA/IIB, T3, IP66. Instruments shall be Intrinsically safe certified (Ex 'i' certified) and	By Supplier

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		Weather proof IP 66 minimum. Ex'd' certification is acceptable in case if intrinsically safe certification is not available. CCOE approval is required.	
9	Necessary structural support & bolts	Angle bar around the dock leveler and Anchor bolts	By supplier
10	Input for civil work	Sizing details of bolt, foundation details (During Detail Engineering)	By Supplier
11	Emergency locking / protection system	Immobilizer security system in the event of emergency stop or power failure. Any braking system if required shall be considered by supplier in their scope of supply / work	By Supplier
12	Bumpers	As per Supplier Recommendation, If required	By Supplier
13	Personnel safeguard	Side mobile toe guards, Visual signs on mobile parts etc	
14	Any other accessories	Any other accessories, equipment which are required but not listed in this specification shall be considered by supplier in their scope	By supplier
15	First Fill of hydraulic oil	Supplier shall consider first fill of hydraulic oil in their scope of supply & furnish the hydraulic oil details ISO viscosity grade.	By Supplier
16	Spare parts	1. Spares required for construction / commissioning 2. Spares required for 2-Years normal operation & maintenance (Priced List) 3. Mandatory Spare as per attached Doc No: IEGP-6351-8840-PH-000-004	By supplier
17	Approved Motor Vendors	Sub Supplier shall be as per attached approved Master supplier list (ref File "MSL-T3")	By supplier

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2. LIST OF ATTACHMENTS

Sl. No.	Document No.	Rev. No.	Title of the Document / Drawing
A	**GENERAL**		
1	PDRP-8440-SP-0013	T1	EQUIPMENT NOISE SPECIFICATION
2	PDRP-8440-SP-0006	F3	PROTECTIVE PAINTING SYSTEMS
3	IEGP-6351-8840-PH-000-004	A1	SPARE PHILOSOPHY
4	IEGP-6351-8610-SL-000-0001	T3	MASTER SUPPLIER LIST
5	IPEG-6389-8310-39-002-0003	I1	ANCHOR BOLT DETAILS
6	IPEG-6389-8310-PH-002-0001	A1	DESIGN BASIS FOR CIVIL, STRUCTURE AND BUILDING WORKS
7	-	-	DOCK LEVELLER LOCATION IN TEG BARRELLING LAYOUT
8	PDRP-8820-SP-0001	F2	BASIC ENGINEERING DESIGN DATA
9	IPEG-6389-8830-QA-002-0006	A1	POSITIVE MATERIAL IDENTIFICATION PROCEDURE
10	IPEG-6389-QA-GN-002	01	INSPECTION COORDINATION PROCEDURE
B	**ELECTRICAL SPECIFICATIONS**		
1	IEGP-6351-8531-SP-000-1001	I2	INDUCTION MOTORS
2	PDRP-8531-SP-0016	F3	ELECTRICAL REQUIREMENT IN PACKAGE SUPPLY EQUIPMENT UNITS
3	PDRP-8531-SP-0010	F6	EARTHING PHILOSOPHY
4	IEGP-6351-8531-TP-000-1001	I2	INSPECTION TEST PLAN FOR INDUCTION MOTOR
5	-	-	DESIGN PHILOSOPHY [MOTOR ADDITIONAL DETAILS]
6	-	-	ELECTRICAL DESIGN BASIS EXTRACT)

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